

Mahdi Sarfarazi

Technical Product Manager · AI & engineering software

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Technical product manager for AI and engineering software, promoted from the engineering team. I took an AI-assisted control-design platform from concept to a validated MVP. Years of hands-on work in robotics, ROS2, control systems and reinforcement learning help me write requirements engineers can build, and judge whether a result is real.

EXPERIENCE

Technical Product Manager · Agentic-AI startup (name withheld) Mar 2026 to present

- Took an AI-assisted control-design platform from concept to MVP. It completed two proof-of-concept cycles (the first with 38 academic test users and 198 projects) and passed the internal product-readiness gate.
- Wrote the MRD, business plan and business model canvas. Ran market research, competitive analysis and more than 100 customer interviews, and turned the findings into requirements and a prioritized roadmap.
- Own the backlog, user stories and acceptance criteria. Coordinate 7 engineers and 3 stakeholders across AI, control engineering, software, design and QA, and check every feature against its acceptance criteria before release.
- Wrote the go-to-market strategy and plan for the next phase, a formal B2B product-market-fit programme with a staged launch and criteria for company-level evidence.

Robotics Software Engineer · same startup Nov 2025 to Mar 2026

- Designed an agent workflow that turns nonlinear system descriptions into operating points, state-space models, controller candidates and closed-loop simulations, and added linearization, stability and frequency-domain checks to its automated verification.

Product and Technical Lead, ViGenX · independent open-source project 2025 to present

- Defined and built an AI video editor that compiles a plain-language brief into a typed workflow you can edit. The LLM only plans, and nothing renders until a person approves it.
- Showed the initial MVP at Diginext's 10th Startup Camp. The public Apache-2.0 repository has CI and tests and is pre-1.0.

Researcher, Robotics Software · AI & Dynamic Systems Lab, Iran University of Science and Technology Nov 2024 to present

- Define requirements, evaluation criteria and success metrics for RL locomotion and risk-aware navigation. Co-authored research on hierarchical deep RL for quadruped navigation that reached 94.2% success and 71.4% fewer collisions than a monolithic baseline in Isaac Gym simulation (ICRoM 2025).

Robotics Engineer and Research Assistant · FUM Center of Advanced Rehabilitation and Robotics Research 2022 to Aug 2024

- Project manager and first author for the FUM-NEXA knee exoskeleton's real-time fuzzy assist-as-needed controller on ROS2 and CAN, which cut muscle effort by 30% at slow sit-to-stand in an engineering study.
- Built modular ROS2 C++/Python stacks for exoskeleton sensing, motor control and calibration. The FUM-HEXA team won the Iraj Yazdanbakhsh First Prize.

Freelance robotics and control engineer (Karlancer, Parscoders · 2024 to 2025): control-system design, neural-network modelling and CAD for clients.

Digital marketing (2018 to 2022): SEO, email and campaign roles, where I first learned about acquisition, funnels and user journeys.

EDUCATION AND TRAINING

M.Sc. Mechatronics Engineering (in progress), Iran University of Science and Technology, 2024 to present · thesis in progress on risk-aware quadruped navigation · Bronze Medal, 2025 Talent Development Award

B.Sc. Mechanical Engineering, Ferdowsi University of Mashhad, 2018 to 2024 · Best Technologist Student of 2023

Product training: Digital Product Management: Modern Fundamentals, University of Virginia (2026) · Diploma in Product Management, Alison (2024) · Foundations of Project Management, Google (2024)

PUBLICATIONS

First author, knee-exoskeleton assist-as-needed control, SSRN preprint (2026, not peer-reviewed) · second author, hierarchical DRL quadruped navigation, ICRoM 2025 · co-author, neural-network intent estimation for exoskeleton users, ICRoM 2022 (IEEE).

SKILLS

Product: discovery and customer interviews, MRD/PRD, roadmapping and prioritization, user stories and acceptance criteria, release validation, POC design, go-to-market planning, Agile/Scrum.

Robotics and control: ROS2, Python, C++, Isaac Gym/Sim, Gazebo, MoveIt, MATLAB/Simulink, PID, fuzzy, sliding-mode, MPC/MPPI, sim-to-real.

AI: LLM and multi-agent systems, constrained planning, deep and distributional RL, computer vision.

Tools and languages: Jira, Notion, Git, Docker, Linux · Persian (native), English (professional).